



Validation and invariance of the Conceptions of Assessment-III Abridged (COA-IIIa) among pre-service and in-service teachers in the United States

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Received: 7 January 2022 / Accepted: 9 April 2023

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Abstract

As beliefs are well-known antecedents of teachers' practices, including assessment practices, sound measurement of teacher beliefs is critical for scholarly research as well as practical purposes. The present study examined the validity of inferences derived from the Conceptions of Assessment III—Abridged (COA-IIIa) instrument with US PK-12 pre-service ($N=554$) and in-service ($N=341$) teachers. Prior research on the COA-IIIa in diverse contexts and with different teacher populations has yielded highly divergent findings concerning this instrument's internal score structure. The study relies on confirmatory factor analyses to investigate the COA-IIIa's score structure among US pre-service and in-service teachers and to test the measurement invariance of select COA-IIIa measurement models across these groups. Findings indicate poor fit of most prior COA-IIIa measurement models with US pre-service and in-service teachers, though we were able to specify several simplified models that were at least metrically invariant across these populations.

Keywords Assessment · Beliefs · In-service teachers · Pre-service teachers · Confirmatory factor analysis · Invariance

Educational assessment—the systematic collection, synthesis, and interpretation of data to inform decisions—is a key facet of education systems globally (Mullis et al., 2012). Teachers are important actors in educational assessment processes, especially those occurring within schools and classrooms. Teachers are expected to be able to design and develop (or select from available) assessments and administer them under appropriate conditions for students. Moreover, teachers need to analyze and interpret data from their own classroom assessments, as well as external assessments, and translate those interpretations into action (Marsh, 2012; Pellegrino et al., 2001).

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The literature evidences a key role of teacher beliefs in the implementation of assessment practices and policies (Brown, 2004; Datnow & Hubbard, 2015). Reeves et al. (2016), for example, found that the frequency of in-service teachers' use of data for instructional decision-making was a function of teacher beliefs about assessment and their own efficacy, not several other factors. At the same time, Kerr et al. (2006) found that teacher beliefs about assessment data invalidity can undermine initiatives intended to promote data-driven decision-making. Teacher beliefs about assessment can also constrain assessment policy implementation (Barnes et al., 2015; Brown, 2004).

While teacher beliefs about assessment are important antecedents of assessment practice (Barnes et al., 2015; Daniels & Poth, 2017), the field's capacity to well measure such beliefs among US pre-service and in-service teachers remains limited. The Conceptions of Assessment III—Abridged (COA-IIIa; Brown, 2006), a measure of teacher beliefs about assessment that is popular globally, has never been fully validated among either US pre-service or in-service teachers. Consequently, the purpose of the present study was to conduct a psychometric evaluation of the COA-IIIa among these populations. Specifically, the present study set out to answer the following research questions vis-à-vis COA-IIIa validity and reliability:

1. What is the COA-IIIa's factor structure among US PK-12 pre-service and in-service teachers, and is the factor structure invariant across these populations?
2. To what extent are COA-IIIa scores reliable among US PK-12 pre-service and in-service teachers?

Proper measurement of teacher beliefs about assessment should prove helpful in the context of research and practice related to assessment policy and practice. For instance, teacher educators might be interested in identifying teachers' current beliefs about assessment, especially negative beliefs, to promote beliefs that are more coherent with the current emphases of education policy and practice (Barnes et al., 2015; Brown, 2004, 2006). Similarly, researchers might be interested in measurement of teacher assessment beliefs in the context of research on assessment policy implementation (Brown, 2004, 2006), the efficacy of data use interventions (Prenger & Schildkamp, 2018), or the scope and nature of teachers' classroom assessment practices (Barnes et al., 2015). Examination of the COA-IIIa's internal score structure among US in-service and pre-service teachers may also yield insights about the underlying theoretical structure of assessment-related beliefs in these specific populations.

1 Theoretical framework

1.1 Teacher beliefs

The present study is grounded in research and theory concerning teacher beliefs, in general, and teacher beliefs about assessment, in particular. Pajares (1992, p. 316) defined teacher beliefs as “an individual's judgment[s] of the truth or falsity of ... proposition[s].” Teacher beliefs can theoretically exert impact on teachers'

instructional behaviors or practices, as well as how teachers interpret information such as policies (Barnes et al., 2015; Clark & Peterson, 1986; Pajares, 1992; Tschannen-Moran et al., 1998). As such, teacher beliefs can indirectly influence student outcomes and the success of policy initiatives.

Research also supports the notion that teacher beliefs about assessment can affect teachers' assessment practices and behaviors specifically (Barnes et al., 2015; Datnow & Hubbard, 2015; Harris & Brown, 2009; Prenger & Schildkamp, 2018). For example, the belief that assessment data are invalid has been shown to serve as a barrier to data use by teachers (Kerr et al., 2006). Similarly, Reeves et al. (2016) found that data use practices were more frequent among teachers who held the belief that assessment improves teaching, and Daniels and Poth (2017) found that teacher beliefs about assessment were associated with their intended approaches to assessment.

The universe of potential teacher beliefs about assessment is large and varied (Barnes et al., 2015; Harris & Brown, 2009). Relevant to the present study, Brown (2004) posited several distinct beliefs teachers could hold concerning the purpose of assessment. These beliefs were that *Assessment Improves Education*, *Assessment Makes Students Accountable*, *Assessment Makes Schools Accountable*, and *Assessment Is Irrelevant*. Some of these general beliefs about assessment were understood to subsume more specific sets of beliefs. As teacher beliefs are multiply determined by individual, cultural, and/or organizational factors, however, the structure and nature of teachers' beliefs about assessment can vary within and/or across contexts (Barnes et al., 2015; Datnow & Hubbard, 2015).

1.2 Prior research on the COA-IIIa and in-service teachers

The COA-IIIa (Brown, 2006) instrument is a 27-item self-report measure designed to support inferences concerning teachers' beliefs about assessment. The COA-IIIa is a short form of the earlier, 50-item Conceptions of Assessment—III instrument (COA-III; Brown, 2004). Like its predecessor, the *COA-IIIa* is intended to measure four general teacher beliefs about the purpose of assessment: *Assessment Makes Schools Accountable*, *Assessment Makes Students Accountable*, *Assessment Improves Education*, and *Assessment Is Irrelevant*. The latter two general beliefs (*Assessment Improves Education* and *Assessment Is Irrelevant*) are hypothesized to each subsume several specific teacher beliefs, specifically *Assessment Improves Teaching*, *Assessment Improves Learning*, *Assessment Is Valid*, *Assessment Describes Abilities*, *Assessment Is Bad*, *Assessment Is Ignored*, and *Assessment Is Inaccurate* (see Fig. 1).

Using confirmatory factor analysis, Brown (2004) found that the hypothesized factor structure well accounted for COA-III data in a large sample of primary in-service teachers from New Zealand and Queensland. Analyses of *COA-IIIa* data in samples of both New Zealand and Queensland primary in-service teachers also showed good model data fit and similar inter-factor correlations relative to the full COA-III. However, the sizes of the *COA-IIIa* inter-factor correlations differed somewhat across the New Zealand and Queensland teacher samples (Brown, 2006).

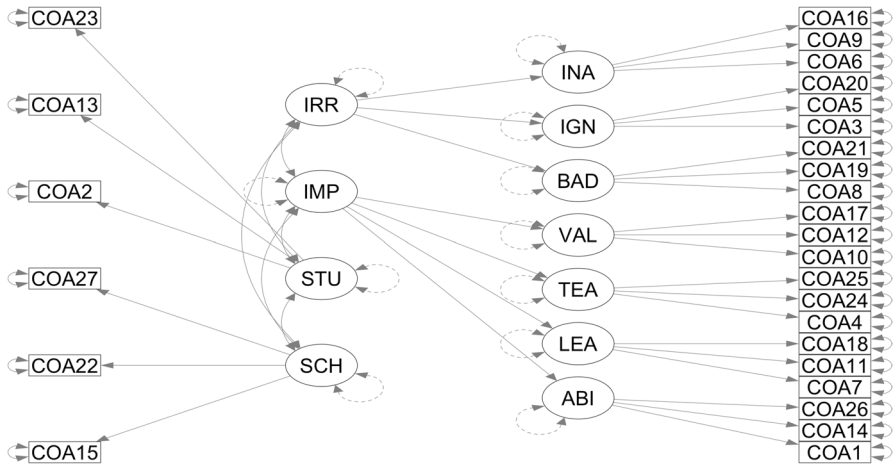


Fig. 1 Graphical representation of Brown's (2006) COA-III model. The model featured four meta-constructs: *Assessment Makes Schools Accountable* (SCH), *Assessment Makes Students Accountable* (STU), *Assessment Improves Education* (IMP), and *Assessment Is Irrelevant* (IRR). IMP subsumes four latent variables: *Assessment Describes Abilities* (ABI), *Assessment Improves Learning* (LEA), *Assessment Improves Teaching* (TEA), and *Assessment Is Valid* (VAL). IRR subsumes three latent variables: *Assessment Is Bad* (BAD), *Assessment Is Ignored* (IGN), and *Assessment Is Inaccurate* (INA). Three items each load directly on SCH and STU

Much subsequent research has examined the COA-III's internal score structure among in-service teacher populations internationally.¹ However, research studies from Hong Kong (Brown et al., 2009), the Netherlands (Segers & Tillema, 2011), Queensland, and Cyprus have all identified different factor structures in terms of the number, hierarchical structure, and organization of the various belief factors.

Given the cross-national heterogeneity observed in the COA's factor structure, Brown et al. (2019) recently tested the generalizability of various models. In the most comprehensive COA-III study to date, the authors tested the measurement invariance of eight measurement models across 11 datasets gathered from various contexts. Only one model—the so-called India model from Brown et al. (2015)—exhibited configural invariance across contexts, though more rigorous tests of model parameter equivalence indicated non-invariance. The India model featured three major assessment belief factors indexed by 21 of the original 27 COA-III items. Their findings suggested there may be some universality cross-nationally as it pertains to these three beliefs.

¹ Our review of the literature focuses only on peer-reviewed studies in which the COA-III was unmodified in terms of item content or response scale (to the extent we were able to verify this). However, some studies reviewed did translate the language of the items.

While psychometric evidence related to the COA-IIIa internationally abounds, very little is known about the performance of the instrument with US in-service teachers. Only one published study reported evidence of the internal score structure of the COA-IIIa among US in-service teachers. In their small sample of K-12 in-service teachers, Barnes et al. (2017) identified via exploratory factor analysis a three-factor model comprising factors they named as follows: *Assessment as Irrelevant*, *Assessment as Valid for Accountability*, and *Assessment Improves Teaching and Learning*.

1.3 Prior research on the COA-IIIa and pre-service teachers

Less published research has addressed the COA-IIIa's factorial structure among pre-service teachers, even though it might conceivably be (and sometimes is) used with this population. Validating the COA-IIIa's factor structure with pre-service teachers may be especially important given potential differences in their levels of experience and expertise—and thus assessment belief structures—relative to in-service teachers (Brown & Remesal, 2012; Daniels et al., 2014). Just as the underlying assessment belief structure may vary among teachers cross-nationally or across school levels (Brown et al., 2019), it may vary between in-service and pre-service teacher populations.

In general, prior structural validity studies conducted internationally have not provided evidence that the originally hypothesized COA-III model fits among pre-service teachers. This was the case in both Brown and Remesal's (2012) study with Spanish pre-service teachers and Gebriel and Brown's (2014) study with Egyptian pre-service teachers. In these studies, the authors were able to specify better-fitting models, but this involved making changes to the hierarchical model structure or which items load on particular factors.

Daniels et al. (2014) also explored the COA-IIIa's factor structure among Canadian pre-service teachers. The authors systematically tested the fit of six different models among these pre-service teachers using confirmatory factor analysis. Ultimately, the model that fit best comprised nine first-order factors only, each comprising two to three manifest variables. The authors removed some items (COA9, "Assessment is an imprecise process" from the *Assessment Is Inaccurate* factor; COA2, "Assessment places students into categories" from the *Assessment Makes Students Accountable* factor; and COA14, "Assessment measures students' higher-order thinking skills" from the *Assessment Describes Abilities* factor) for conceptual or empirical reasons.

Finally, a conference paper by Kraner and Reeves (2018) examined the COA-IIIa's factor structure among US pre-service teachers. The authors compared the fit of four models using confirmatory factor analysis. Only one model was acceptable, which featured 24 items and eight non-hierarchical factors, eliminating the *Assessment Makes Students Accountable* factor. Two of the original items from the *Assessment Makes Students Accountable* factor (COA2, "Assessment places students into categories," and COA23, "Assessment is assigning a grade or level to student work") had been removed based on the findings of Daniels et al. (2014), and the remaining

item (COA13, “Assessment determines if students meet qualifications standards”) was reclassified on the existing *Assessment Is Valid* factor. In addition, COA9 (“Assessment is an imprecise process”) was removed from the *Assessment Is Inaccurate* factor.

2 The present study

Prior studies clearly indicate cross-national variation in the COA-IIIa’s factor structure with both in-service and pre-service teachers, which is interesting from not only methodological but also substantive perspectives. However, this international body of literature base features no thorough psychometric evaluation of the COA-IIIa among US in-service and especially pre-service teachers. Indeed, there have also been calls to examine COA-IIIa reliability and validity issues in contexts with higher stakes associated with assessment such as the US (Brown, 2011; Brown & Remesal, 2012; Brown et al., 2019).

The central aim of the present study, then, is to attempt to validate the COA-IIIa instrument—vis-à-vis its internal score structure—with US pre-service and in-service teachers. We follow the approach of several published COA-IIIa studies that compared the fit of various COA-IIIa factor models. These include prior studies by Daniels et al. (2014), Brown and Remesal (2012), Brown et al. (2019), and Kraner and Reeves (2018). Table 1 summarizes the 16 models we tested a priori among US pre-service and in-service teachers, and the relationship between these models tested and prior studies. In addition to testing independently the fit of these models among US pre-service and in-service teachers, we also formally investigated the invariance of select COA-IIIa structures.

3 Methods

3.1 Participants

Participants were US PK-12 pre-service ($N=554$) and in-service ($N=341$) teachers. Recruitment of pre-service teachers occurred from Spring 2014 to Spring 2017 and relied on several mechanisms. First, individuals were recruited from undergraduate pre-service teacher education courses at the authors’ Midwestern regional-comprehensive university. Second, individuals within a sampling frame maintained by the investigator were invited to participate in this research. Third, recruitment occurred via social media outlets such as Facebook, Reddit, and Twitter. We also leveraged data gathered in the context of several other studies conducted by the research team’s members, including survey research and intervention studies. Some of the intervention studies featured both pretest and post-test data collections, and when data from both time points were available, the relevant pretest data were used here.

This study’s pre-service teacher (PST) participants were pre-K-12 PSTs enrolled in one of at least 75 traditional teacher preparation programs (89.58% undergraduate

Table 1 Summary of models tested for US pre-service and in-service teachers

Model number	Prior study				
	Daniels et al. (2014)	Brown and Remesal (2012)	Brown et al. (2019)	Kraner and Reeves (2018)	Barnes et al. (2017)
1	Model 1	Model A	New Zealand model (Brown, 2006)	Model 1	—
2	Model 2	—	—	Model 2	—
3	Model 3	—	—	Model 3	—
4	Model 4	—	Cyprus model (Brown & Michaelides, 2011)	—	—
5	Model 5	—	—	—	—
6	Model 6	—	—	—	—
7	—	Model B	—	—	—
8	—	Model C	Catalonia model (Brown & Remesal, 2012)	—	—
9	—	Model D	—	—	—
10	—	—	—	Model 4	—
11	—	—	Queensland model (Brown et al., 2011)	—	—
12	—	—	Hong Kong model (Brown et al., 2009)	—	—
13	—	—	Egypt model (Gebril & Brown, 2014)	—	—
14	—	—	India model (Brown et al., 2015)	—	—
15	—	—	Ecuador model (Brown & Remesal, 2017)	—	—
16	—	—	—	—	Barnes et al.'s (2017) model

For each model tested in our study, Table 1 shows how prior studies have also tested these models (and if applicable notes the number, label, or name of the model in those studies)

program, 8.33% graduate program, and 2.08% “other”² program level) in at least 31 states. Based on available data, the sample was 88.43% female, 86.59% white, and 10.33% reported being of Hispanic/Latino ethnicity. The PST participants’ mean age was 24.93 (SD=6.85). The PST sample represented future teachers of elementary (K-5) (73.73%), middle (6–8) (9.68%), and high (9–12) (9.22%) school grades, as well as teachers of pre-kindergarten (0.46%) and of multiple school levels (5.99%).³ While some PST participants reported intending to be subject-specialist teachers (e.g., mathematics, English/language arts, music), most (66.36%) reported intending to teach all subject areas (i.e., elementary teachers). Most individuals within the PST sample (81.57%) were concurrently participating in a clinical or field experience at the time of data collection. The average number of years in which the PSTs had been enrolled in a teacher preparation program was 2.75 (SD=1.11).

The in-service teacher (IST) data were collected in the context of several other studies conducted by the research team but are limited to data collected from Illinois teachers. The IST participants were teachers working in at least 55 districts (7.72% urban district, 77.81% suburban, 8.04% town, and 6.43% rural) and at least 73 schools. The IST sample represented teachers of elementary (K-5) (32.02%), middle (6–8) (39.93%), and high (9–12) (22.66%) school grades, as well as teachers of pre-kindergarten (1.08%) and of multiple school levels (2.88%).⁴ The IST sample was 78.80% female. Of the ISTs, 97.67% reported being white, and 2.66% reported being Hispanic/Latino. The IST participants’ mean age was 40.74 (SD=11.27). The IST sample was heterogeneous with respect to years of experience, which ranged from 1 to 45 years ($M=13.38$, $SD=8.98$).⁵ The sizes of both the PST and IST samples were deemed adequate based on evidence available from prior simulation studies related to factor analytic techniques and invariance testing therein (Cheung & Rensvold, 2002; MacCallum et al., 1999; Meade & Bauer, 2007). This study was approved by the institutional review board at the authors’ institution (Protocol #: HS17-0112). The datasets generated during and/or analyzed during the current study are available from the corresponding author on reasonable request.

3.2 Instrumentation

The primary data source was the 27-item COA-III instrument (Brown, 2006). As noted, the COA-III was designed to measure various hierarchically structured teacher beliefs about assessment (e.g., *Assessment Is Valid*, *Assessment Makes Students Accountable*, *Assessment Is Bad*). In its original design, each unique

² This included post-baccalaureate programs and combined undergraduate and graduate teacher education programs.

³ Of the PST sample, 0.92% selected “other” for their intended school level.

⁴ School level taught was indeterminable for 1.44% of IST participants with a non-missing response for this variable.

⁵ Missing data for PST demographic and professional variables constituted between 0.18 and 72.02% of PST cases and between 0.29 and 18.48% of IST cases. Some of this is due to inconsistencies across studies and datasets in available non-COA-III data. Some of these estimates should be considered very unreliable.

assessment belief is operationalized by three items. The response format for the 27 COA-III items was a 6-point, “positively-packed” rating scale: 1 = *Strongly disagree*, 2 = *Mostly disagree*, 3 = *Slightly agree*, 4 = *Moderately agree*, 5 = *Mostly agree*, and 6 = *Strongly agree*. As discussed earlier, the number of items and factors and the organization of factors that best account for responses to the instrument have varied in prior research. COA-III item content is shown in Table 2 under Brown’s (2006) originally model configuration.

3.3 Analytic approach

All analyses were conducted in R and within a confirmatory factor analytic framework. The number of missing item-level COA-III observations was negligible, ranging between 0 (0% of all cases) and 9 (1.01%). Little’s (1988) missing completely at random (MCAR) test was not statistically significant ($\chi^2(652) = 633.32$, $p = 0.69$). However, Mardia’s (1970) multivariate skewness ($b_1 = 8523.58$, $p < 0.001$) and kurtosis ($b_2 = 54.67$, $p < 0.001$) tests were both statistically significant.

Given the presence of missing data and the untenability of the maximum likelihood estimator’s assumption of multivariate normality, we deployed a robust, full-information estimator (Olsson et al., 2000), specifically “MLR” in R’s lavaan package (Rosseel, 2012). For scaling purposes, the residual variances of the latent variables were fixed to one in all analyses. We allowed all exogenous latent variables to covary in all analyses. We did this to provide the fairest possible comparison of factor structures, in recognition of the fact that additional covariance parameters can improve fit.

In accord with best practices, we used chi-square goodness-of-fit tests in conjunction with other fit indices such as the comparative fit index (CFI), the standardized root mean square residual (SRMR), and root mean square error of approximation (RMSEA) to examine model fit (Bentler, 1990; Bentler & Bonett, 1980; Hu & Bentler, 1999; Steiger & Lind, 1980). Given that CFI may be depressed in multi-dimensional models, we emphasized the SRMR and gamma hat indices in evaluating models (Fan & Sivo, 2007). For each model, our evaluation process generally involved inspection of fit tests and indices and factor correlations at the overall model level, and communalities and factor loadings at the item level.

For select models, we formally tested the invariance of model parameters across US in-service and pre-service teachers using multi-group confirmatory factor analysis (CFA) (Meredith, 1993). This involved comparing the fit of successive multi-group CFA analyses in which parameters were increasingly constrained equal across groups. Invariance testing began by fitting a multi-group CFA model in which parameters were unconstrained across groups (model 0), followed by a fitting series of multi-group CFA models with increasing equality constraints. These subsequent models featured equal factor loadings across groups (model 1); equal factor loadings and item intercepts across groups (model 2); equal factor loadings, item intercepts, and residual item variances/covariances across groups (model 3); and, finally, equal item factor loadings, item intercepts, residual item variances/covariances, and factor variances/covariances across groups (model 4; Dimitrov, 2010).

Table 2 Descriptive statistics for all items for US pre-service and in-service teachers

Item number	Item content	Factor	PST			IST				
			M	SD	S _i	K _i	M	SD	S _i	K _i
COA15	Assessment provides information on how well schools are doing	Assessment Makes Schools Accountable	3.50	1.16	0.07	2.68	3.22	1.17	0.01	2.33
COA22	Assessment is an accurate indicator of a school's quality	Assessment Makes Schools Accountable	2.71	1.03	0.55	3.26	2.64	1.07	0.29	2.54
COA27	Assessment is a good way to evaluate a school	Assessment Makes Schools Accountable	2.87	1.15	0.30	2.68	2.74	1.19	0.35	2.65
COA2	Assessment places students into categories	Assessment Makes Students Accountable	4.20	1.12	-0.44	2.96	4.01	1.12	-0.47	2.90
COA13	Assessment determines if students meet qualifications standards	Assessment Makes Students Accountable	3.93	1.09	-0.019	2.60	3.68	1.07	-0.19	2.67
COA23	Assessment is assigning a grade or level to student work	Assessment Makes Students Accountable	2.96	1.25	0.30	2.50	2.88	1.24	0.38	2.48
COA26	Assessment is a way to determine how much students have learned from teaching	Assessment Improves Education; Assessment Describes Abilities	4.19	1.17	-0.37	2.69	4.02	1.15	-0.30	2.64
COA1	Assessment establishes what students have learned	Assessment Improves Education; Assessment Describes Abilities	4.50	1.09	-0.77	3.39	4.31	0.98	-0.42	2.86
COA14	Assessment measures students' higher-order thinking skills	Assessment Improves Education; Assessment Describes Abilities	3.50	1.18	-0.04	2.61	3.22	1.19	0.03	2.27
COA11	Assessment provides feedback to students about their performance	Assessment Improves Education; Assessment Improves Learning	4.19	1.13	-0.40	2.61	4.02	1.16	-0.29	2.48
COA7	Assessment feeds back to students their learning needs	Assessment Improves Education; Assessment Improves Learning	4.17	1.19	-0.31	2.35	3.85	1.27	-0.21	2.34
COA18	Assessment helps students improve their learning	Assessment Improves Education; Assessment Improves Learning	3.82	1.16	-0.15	2.58	3.48	1.17	-0.04	2.57
COA25	Assessment is integrated with teaching practice	Assessment Improves Education; Assessment Improves Teaching	4.27	1.16	-0.35	2.62	4.12	1.16	-0.33	2.72
COA4	Assessment information modifies ongoing teaching of students	Assessment Improves Education; Assessment Improves Teaching	4.72	1.08	-0.68	3.07	4.32	1.14	-0.41	2.69

Table 2 (continued)

Item number	Item content	Factor	PST			IST				
			M	SD	S _i	K _i	M	SD	S _i	K _i
COA24	Assessment allows different students to get different instructions	Assessment Improves Education; Assessment Improves Teaching	3.73	1.30	-0.13	2.29	3.62	1.25	0.00	2.34
COA10	Assessment results are trustworthy	Assessment Improves Education; Assessment Is Valid	3.57	1.05	-0.18	2.64	3.50	1.06	-0.04	2.55
COA17	Assessment results are consistent	Assessment Improves Education; Assessment Is Valid	3.02	1.08	0.20	3.05	3.12	1.08	-0.00	2.48
COA12	Assessment results can be depended on	Assessment Improves Education; Assessment Is Valid	3.72	1.03	-0.05	2.67	3.54	1.05	-0.21	2.63
COA21	Assessment forces teachers to teach in a way against their beliefs	Assessment Is Irrelevant; Assessment Is Bad	2.70	1.28	0.75	2.98	2.90	1.27	0.63	2.85
COA19	Assessment is unfair to students	Assessment Is Irrelevant; Assessment Is Bad	2.69	1.02	0.80	3.59	2.68	1.16	0.87	3.54
COA8	Assessment interferes with teaching	Assessment Is Irrelevant; Assessment Is Bad	2.40	1.16	0.91	3.59	3.06	1.36	0.40	2.54
COA20	Teachers conduct assessments but make little use of the results	Assessment Is Irrelevant; Assessment Is Ignored	2.85	1.13	0.50	3.05	3.04	1.25	0.54	2.72
COA3	Assessment results are filed and ignored	Assessment Is Irrelevant; Assessment Is Ignored	2.34	1.06	0.94	3.88	2.77	1.38	0.58	2.44
COA5	Assessment has little impact on teaching	Assessment Is Irrelevant; Assessment Is Ignored	1.77	0.96	1.91	7.68	2.10	1.12	1.13	3.82
COA16	Assessment results should be treated cautiously given measurement error	Assessment Is Irrelevant; Assessment Is Inaccurate	4.42	1.17	-0.14	1.98	4.30	1.11	-0.25	2.51
COA6	Teachers should take into account the error and imprecision in all assessment	Assessment Is Irrelevant; Assessment Is Inaccurate	4.77	1.23	-0.76	2.69	4.28	1.20	-0.22	2.23
COA9	Assessment is an imprecise process	Assessment Is Irrelevant; Assessment Is Inaccurate	2.80	1.20	0.81	3.32	3.31	1.29	0.40	2.47

Items are designated according to Brown's (2006) original model structure

PST pre-service teacher, *IST* in-service teacher

We relied on both likelihood-ratio tests of goodness-of-fit and changes in the CFI in evaluating invariance across groups. Given that chi-square tests are especially sensitive with large samples, we emphasized the CFI with reductions of less than 0.01 in this index considered indicative of invariance. We also emphasized gamma hat with reductions of less than 0.001 in this index considered indicative of invariance (Cheung & Rensvold, 2002).

We also examined the direction and magnitudes of correlations among the latent factors. We finally assessed score reliability for the final model using coefficient omegas (ω ; McDonald, 2013) for all factors. Moreover, assuming tau equivalence and given that COA-III A users may average subscale item scores (i.e., weighting the items equally) to form composite variables, we used Cronbach's alpha to estimate internal score consistency. Finally, we computed construct replicability for all factors (H ; Hancock & Mueller, 2001) using Hammer's (2016) Excel calculator.

4 Results

4.1 Single-group CFA and model refinement

Table 2 contains descriptive statistics for all items in the PST and IST samples. Table 3 then summarizes the fit of the 16 potential factor models when estimated in single-group CFAs with PSTs and ISTs separately. Given the large number of models, the aim of these single-group CFAs was to identify well-fitting models for which formal invariance testing across PSTs and ISTs may be warranted.

As seen in Table 3, some of the models or fit statistics pertaining thereto were inestimable in one or both groups due to non-convergence or a non-positive definite matrix. These inestimable models included the original "New Zealand" model (Brown, 2006). The inestimability of these models may indicate excessive model complexity, poor model fit, or other issues.

Of those models which were estimable, initial interpretation of results emphasized global model fit. Three models (5, 6, and 10) had the best global fit vis-à-vis the CFI, SRMR, and gamma hat indices. These three models were then examined in relation to more local measures (e.g., factor loadings, communalities) and from substantive perspectives. This closer review suggested that, as specified, models 5 and 6 (both from Daniels et al.'s (2014) Canadian PST study) and model 10 (from Kraner and Reeves's (2018) PST study) may each feature structural and/or substantive problems. In what follows, we discuss these issues with these models and how we derived further models therefrom to address their issues.

4.1.1 Model 5

Model 5 (Daniels et al., 2014) featured 24 items and all nine first-order factors from Brown (2006), though there were no second-order factors. However, three items were excluded: first, COA2 ("Assessment places students into categories") from the *Assessment Makes Students Accountable* factor; second, COA9 ("Assessment is an imprecise process") from the *Assessment Is Inaccurate* factor; and third, COA14

(“Assessment measures students’ higher-order thinking skills”) from the *Assessment Describes Abilities* factor.

As noted, global fit of model 5 was the best for ISTs, and better for ISTs than PSTs. However, for both PSTs and ISTs, the two items on the *Assessment Makes Students Accountable* factor featured negligibly small factor loadings: COA13 (“Assessment determines if students meet qualifications standards”) and COA23 (“Assessment is assigning a grade or level to student work”). In addition, substantive analysis of item content for COA13 raised questions about whether “qualifications standards” would be well understood by US educators. This analysis thus indicated possible measurement and/or substantive issues for the *Assessment Makes Students Accountable* factor.

4.1.2 Model 5b

Before abandoning model 5 entirely, we derived a model which removed the *Assessment Makes Students Accountable* factor and its remaining two items entirely. This ameliorated the issue from model 5 in which the two items loading on this factor were negligible, as well as the substantive issue with item COA13 (“Assessment determines if students meet qualifications standards”). Model 5b then featured 22 items and 8 first-order factors from Brown (2006), excluding the *Assessment Makes Students Accountable* factor. As show in Table 3, the global fit of model 5b was much improved relative to model 5 for both ISTs and PSTs and fit the ISTs especially well. We subsequently used model 5b as a basis for invariance testing.

4.1.3 Model 6

Model 6 (Daniels et al., 2014) featured 22 items and seven first-order factors and no second-order factors. Model 6 had featured the new factor *Assessment Is Reporting Achievement* manifested by COA13 (“Assessment determines if students meet qualifications standards”) and COA23 (“Assessment is assigning a grade or level to student work”) from the *Assessment Makes Students Accountable* factor and two other items (COA1, “Assessment establishes what students have learned,” and COA26, “Assessment is a way to determine how much students have learned from teaching,” both originally from the *Assessment Describes Abilities* factor). In other words, Daniels et al. (2014) had combined the two remaining items from both the *Assessment Makes Students Accountable* and *Assessment Describes Abilities* factors from their model 5. Model 6 also excluded the *Assessment Is Inaccurate* factor and its two items.

For both ISTs and PSTs, while the two items from the *Assessment Describes Abilities* factor and COA13 (“Assessment determines if students meet qualifications standards”) from the *Assessment Makes Students Accountable* factor loaded adequately well on *Assessment Is Reporting Achievement*, the loading for COA23 (“Assessment is assigning a grade or level to student work”) was negligible. Given this observation as well as to-be-noted substantive issues with this model, two other models were derived from this model and subsequently tested.

Table 3 Single-group model fit for US pre-service and in-service teachers

Model number	Population	χ^2 ^a	df	χ^2	CFI ^b	CFI ^c	RMSEA (90% CI) ^b	RMSEA (90% CI) ^c	SRMR	Gamma hat
1	PST	NA	NA	NA	NA	NA	NA	NA	NA	NA
2	PST	1317.97***	318	0.785	0.791	0.791	0.079 (0.072–0.080)	0.081 (0.076–0.080)	0.084	0.86
3	PST	1006.77***	288	0.845	0.852	0.852	0.068 (0.064–0.072)	0.072 (0.067–0.076)	0.086	0.90
4	PST	NA	NA	NA	NA	NA	NA	NA	NA	NA
5	PST	647.76***	216	0.893	0.863	0.863	0.061 (0.056–0.066)	0.064 (0.058–0.070)	0.061	0.93
5b	PST	510.68***	181	0.910	0.916	0.916	0.058 (0.052–0.064)	0.061 (0.054–0.067)	0.054	0.94
6	PST	597.56***	188	0.891	0.896	0.896	0.063 (0.058–0.069)	0.067 (0.061–0.074)	0.062	0.92
6b	PST	502.10***	168	0.909	0.914	0.914	0.061 (0.055–0.066)	0.064 (0.058–0.071)	0.054	0.94
6c	PST	420.64***	149	0.920	0.925	0.925	0.058 (0.052–0.064)	0.061 (0.054–0.068)	0.053	0.94
6d	PST	474.23***	168	0.916	0.921	0.921	0.058 (0.052–0.064)	0.061 (0.055–0.068)	0.053	0.94
7	PST	1191.57***	315	0.812	0.818	0.818	0.072 (0.068–0.076)	0.076 (0.071–0.080)	0.094	0.88
8	PST	1175.02***	309	0.814	0.821	0.821	0.072 (0.068–0.076)	0.076 (0.071–0.081)	0.083	0.88
9	PST	NA	NA	NA	NA	NA	NA	NA	NA	NA
10	PST	639.33***	224	0.902	0.907	0.907	0.058 (0.053–0.064)	0.061 (0.056–0.067)	0.054	0.93
10b	PST	574.07***	202	0.906	0.901	0.901	0.058 (0.053–0.064)	0.061 (0.055–0.067)	0.054	0.94
11	PST	NA	NA	NA	NA	NA	NA	NA	NA	NA
12	PST	NA	NA	NA	NA	NA	NA	NA	NA	NA
13	PST	1103.21***	314	0.83	0.84	0.84	0.068 (0.064–0.072)	0.072 (0.068–0.77)	0.093	0.89
14	PST	742.07***	186	0.846	0.852	0.852	0.074 (0.069–0.079)	0.079 (0.073–0.085)	0.065	0.90
15	PST	1094.88***	262	0.805	0.814	0.814	0.077 (0.072–0.081)	0.081 (0.076–0.086)	0.090	0.88
16	PST	1373.75***	296	0.764	0.769	0.769	0.082 (0.078–0.086)	0.087 (0.082–0.092)	0.091	0.85
1	IST	NA	NA	NA	NA	NA	NA	NA	NA	NA
2	IST	835.10***	318	0.815	0.820	0.820	0.073 (0.067–0.078)	0.080 (0.080–0.073)	0.072	0.86
3	IST	616.13***	288	0.883	0.886	0.886	0.061 (0.055–0.067)	0.067 (0.059–0.074)	0.060	0.90

Table 3 (continued)

Model number	Population	χ^2 ^a	df	χ^2	CFT ^b	CFT ^c	RMSEA (90% CI) ^b	RMSEA (90% CI) ^c	SRMR	Gamma hat
4	IST	NA	NA	NA	NA	NA	NA	NA	NA	NA
5	IST	417.12***	216	0.919	0.922	0.922	0.055 (0.048–0.062)	0.060 (0.052–0.069)	0.053	0.93
5b	IST	311.45***	181	0.942	0.944	0.944	0.048 (0.040–0.056)	0.053 (0.043–0.063)	0.048	0.95
6	IST	405.82***	188	0.907	0.910	0.910	0.061 (0.054–0.068)	0.068 (0.059–0.077)	0.054	0.92
6b	IST	366.87***	168	0.914	0.918	0.918	0.061 (0.054–0.069)	0.069 (0.059–0.078)	0.051	0.92
6c	IST	264.58***	149	0.945	0.947	0.947	0.050 (0.041–0.058)	0.056 (0.044–0.066)	0.048	0.95
6d	IST	319.12***	168	0.933	0.935	0.935	0.053 (0.045–0.061)	0.059 (0.049–0.069)	0.050	0.94
7	IST	744.15***	315	0.847	0.851	0.851	0.066 (0.061–0.072)	0.073 (0.066–0.080)	0.068	0.88
8	IST	796.80***	309	0.826	0.831	0.831	0.071 (0.066–0.077)	0.079 (0.072–0.085)	0.074	0.87
9	IST	564.19***	264	0.873	0.879	0.879	0.060 (0.054–0.067)	0.066 (0.059–0.074)	0.066	0.90
10	IST	464.16***	224	0.908	0.911	0.911	0.058 (0.052–0.065)	0.065 (0.056–0.073)	0.051	0.92
10b	IST	367.65***	202	0.930	0.933	0.933	0.051 (0.043–0.058)	0.056 (0.047–0.065)	0.050	0.94
11	IST	NA	NA	NA	NA	NA	NA	NA	NA	NA
12	IST	NA	NA	NA	NA	NA	NA	NA	NA	NA
13	IST	NA	NA	NA	NA	NA	NA	NA	NA	NA
14	IST	571.16***	186	0.835	0.841	0.841	0.081 (0.074–0.088)	0.089 (0.081–0.097)	0.070	0.87
15	IST	738.24***	262	0.812	0.820	0.820	0.076 (0.071–0.082)	0.084 (0.077–0.091)	0.067	0.86
16	IST	829.13***	296	0.802	0.806	0.806	0.076 (0.071–0.082)	0.084 (0.078–0.091)	0.075	0.85

Model 5b, 6b, 6c, and 10b results are included for comparison purposes, even though these models were not specified a priori. NA indicates that a particular model or fit statistics was inestimable (e.g., due to non-convergence or a non-positive definite matrix)

PST pre-service teacher, IST in-service teacher

*** $p < 0.001$

^aChi-square statistic adjusted for non-normality using Satorra and Bentler's (2001) method

^bCalculated using the Satorra and Bentler's (2001) adjusted chi-square statistic

^cCalculated using the methodology outlined in Brosseau-Liard et al. (2012) and/or Brosseau-Liard and Savalei (2014)

4.1.4 Model 6b

Given the negligible loading of COA23 (“Assessment is assigning a grade or level to student work”) on *Assessment Is Reporting Achievement* in model 6, we estimated a new model (model 6b) that was equivalent to model 6, but which excluded COA23. This model had 21 items and seven first-order factors each with three indicator variables; six of the factors were from Brown (2006), and the seventh was a combination of select items from Brown’s *Assessment Makes Students Accountable* factor (2 items) and *Assessment Describes Abilities* factor (1 item). As shown in Table 3, model 6b improved on fit from model 6 and fit both groups approximately equally well. We thus subsequently used model 6b as a basis for invariance testing.

4.1.5 Model 6c

We also derived a second model based on the results of model 6 (model 6c and as shown in Fig. 2). In addition to removing COA23 (“Assessment is assigning a grade or level to student work”) because of its negligible loading on *Assessment Is Reporting Achievement*, it also removed COA13 (“Assessment determines if students meet qualifications standards”) because of substantive concerns about the item in the US context. Model 6c then featured 20 items and 6 first-order factors from Brown (2006), excluding the *Assessment Makes Students Accountable* and *Assessment Is Inaccurate* factors. The two remaining items on Daniels et al.’s (2014) *Assessment Is Reporting Achievement* factor from model 6 were the two items from Brown’s (2006) *Assessment Describes Abilities* factor, so we term this seventh factor *Assessment Describes Abilities*. As shown in Table 3, model 6c improved on fit from both models 6 and 6b, but fit better for ISTs. We subsequently used model 6c as a basis for invariance testing.

4.1.6 Model 6d

Given that models 6, 6b, and 6c had all removed item COA14 (“Assessment measures students’ higher-order thinking skills”) from the *Assessment Describes Abilities* factor based on the findings from only one study (Daniels et al., 2014), we also estimated an additional model, model 6d, with all three original items for this factor. This increased the number of indicators for this factor from 2 to 3, but it was otherwise identical to model 6c. The model comprised 21 items corresponding to six of Brown’s (2006) factors and excluded Brown’s *Assessment Makes Students Accountable* and *Assessment Is Inaccurate* factors. While it did not improve fit relative to model 6c (see Table 3), we did still evaluate its invariance across PSTs and ISTs for comprehensiveness.

4.1.7 Model 10

Model 10 (Kraner & Reeves, 2018) featured 24 items and eight factors from Brown’s (2006) model, but excluded the *Assessment Makes Students Accountable* factor and featured no second-order factors. In model 10, two of the original items from the *Assessment*

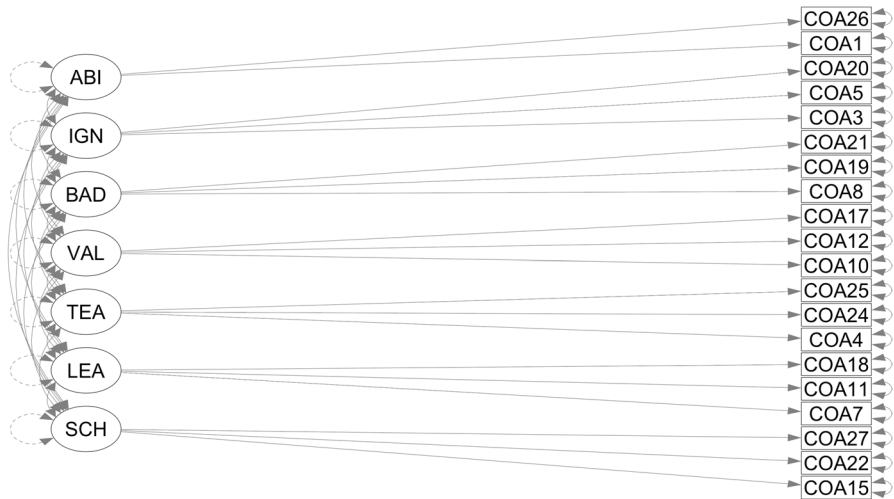


Fig. 2 Graphical representation of model 6c. The model featured seven latent variables: *Assessment Makes Schools Accountable* (SCH), *Assessment Improves Learning* (LEA), *Assessment Improves Teaching* (TEA), *Assessment Is Valid* (VAL), *Assessment Is Bad* (BAD), *Assessment Is Ignored* (IGN), and *Assessment Describes Abilities* (ABI)

Makes Students Accountable factor (COA2, “Assessment places students into categories,” and COA23, “Assessment is assigning a grade or level to student work”) were removed based on the findings of Daniels et al. (2014) and the remaining item (COA13, “Assessment determines if students meet qualifications standards”) was reclassified on the existing *Assessment Is Valid* factor as a fourth manifest variable. In addition, model 10 removed COA9 (“Assessment is an imprecise process”) from the *Assessment Is Inaccurate* factor, as was also done by Daniels et al. (2014). However, model 10 did not remove COA14 (“Assessment measures students’ higher-order thinking skills”) from the *Assessment Describes Abilities* factor as did they.

Global fit for model 10 was generally the best of the initial models for PSTs, and global fit among ISTs and PSTs was most similar. All factor loadings were adequately large and statistically significant, and model 10 did not feature the same problems as in models 5 and 6 (e.g., small factor loadings for some items). Model 10 had addressed these issues by removing COA23 (“Assessment is assigning a grade or level to student work”) and putting COA13 (“Assessment determines if students meet qualifications standards”) on *Assessment Is Valid*.

4.1.8 Model 10b

However, while there was a non-trivial factor loading for COA13 (“Assessment determines if students meet qualifications standards”) on the valid factor in model 10, substantive analysis indicated that this may not have been well justified. COA13 does not

expressly pertain the quality of the information provided by assessment processes. At the same time, “qualifications standards” may be a language that is unfamiliar in the US context. For these reasons, we derived model 10b which did not feature COA13 on the *Assessment Is Valid* factor. Model 10b thus had 23 items and eight single-level factors. It largely preserved the original structure of the COA-IIIa in that it captures 8/9 district factors from Brown (2006) but with no higher-order structure; however, the *Assessment Make Students Accountable* factor was removed, and one item was removed from the *Assessment Is Inaccurate* factor. Single-group CFAs with model 10b in both groups showed that fit generally improved relative to model 10, but this was especially the case for ISTs (see Table 3). We thus tested the invariance of model 10b across PSTs and ISTs.

4.1.9 Other models

When estimating the above models in single-group CFAs, inspection of the matrices of factor inter-correlations revealed high correlations among those factors originally clustered under Brown’s (2006) *Assessment Improves Education* meta-construct. This was particularly the case for *Assessment Describes Abilities* and *Assessment Improves Teaching*, which had a correlation as high as 0.94 in some models when fit with ISTs. Correlations with and among these factors and those other *Assessment Improves Education* sub-dimensions—*Assessment Is Valid* and *Assessment Improves Learning*—were on the order of 0.8. However, these correlations were higher among ISTs.

Obviously, this is highly suggestive that there is a higher-order structure among these factors or that these factors are not distinct. In response to this, we conducted supplemental analyses which re-introduced the *Assessment Improves Education* higher-order factor in all of the candidate models above. Only two of these models (models 5b and 6c with this higher-order factor added) were estimable in both groups. However, neither model fit as well globally as those from which they were derived. We also estimated a derivative of model 6c, the best-fitting model across groups, which organized only *Assessment Describes Abilities* and *Assessment Improves Teaching* under the higher-order factor *Assessment Improves Education*, but global fit again decreased.

These analyses all showed that re-introduction of a higher-order structure did not improve fit. However, the possibility also existed that these highly correlated factors are not actually distinct. We had already tried to fit a model with just four higher-order factors (model 2), and it did not fit well. Given issues with the *Assessment Makes Students Accountable* factor, though, we also tried a derivative model of model 6c comprising the same items but organized into three major factors only, but this did not improve fit relative to model 6c. Finally, we tried a model derived from model 6c with all items from just *Assessment Describes Abilities* and *Assessment Improves Teaching* loading on the same factor, which also did not improve fit.⁶

⁶ As noted by a reviewer, future studies of the COA-IIIa may consider bifactor modeling approaches, which have enjoyed some use in prior research on this instrument (Brown & Remesal, 2017).

4.2 Invariance testing

Table 4 contains results of invariance testing via multi-group CFA for models 5b, 6b, 6c, 6d, and 10b, which were the five best-fitting models overall. For all five models, we established via Δ CFI and Δ gamma hat that the factor loadings can be considered statistically equivalent, that is, the models were metrically equivalent. However, subsequent imposition of the additional constraint that item intercepts were equal across groups showed that scalar invariance was not tenable for any of the models. The degradation in fit as evidenced by the CFI index was larger than 0.01 for all models, and degradation in fit as evidenced by the gamma hat index was larger than 0.001 for all models. While Table 4 contains the results of successively constrained multi-group CFA models that we ran, we do not interpret them here because scalar invariance was not established.

4.3 Factor loadings for final model

Table 5 presents item factor loadings for model 6c, the globally best-fitting model (e.g., per CFI, SRMR, and gamma hat) we tested in both groups. To show how model 6c fits independently for each population, we report these estimates based on single-group CFAs rather than multi-group CFA. As can be seen, nearly all factor loadings were larger than 0.50. One exception was the loading for COA5 (“Assessment has little impact on teaching”) which was 0.46.

4.4 Inter-factor correlations for final model

Table 6 presents factor inter-correlations for model 6c by group. In general, the correlations were in accord with expectations, though there were some differences in these estimates between in-service and pre-service teachers. As noted, there was a very high correlation (0.94) between *Assessment Describes Abilities* and *Assessment Improves Teaching* among ISTs.

4.5 Reliability for final model

Table 7 presents factor-level alpha, omega reliability, and construct replicability estimates for model 6c. Again, as model 6c was not shown to be invariant, we report these estimates by group. In general, the estimates indicated acceptable score reliability levels in both groups for most of the factors. However, reliabilities were slightly lower for the *Assessment Is Ignored* and *Assessment Describes Abilities* factors, especially for PSTs.

5 Discussion

Many educational assessment policies and practices internationally rely heavily on teachers for their successful implementation. Teachers’ beliefs towards assessment are thus important factors to consider given that these beliefs can influence their

Table 4 Summary of invariance test results for US pre-service and in-service teachers

CFA model	χ^2 ^a	df	χ^2 ^a	df	$\Delta\chi^2$	RMSEA (90% CI) ^b	RMSEA (90% CI) ^c	CFI ^b	Δ CFI ^b	CFI ^c	Δ CFI ^c	SRMR	GH	Δ GH
Model 5b invariance testing														
Model 0	810.99***	362	–	–	–	0.054 (0.049–0.058)	0.058 (0.053–0.063)	0.923	–	0.927	–	0.050	0.942	–
Model 1	823.69***	376	14.46	14	14	0.053 (0.048–0.057)	0.057 (0.052–0.062)	0.924	0.001	0.927	0.000	0.052	0.942	0.000
Model 2	928.77***	390	117.93***	14	103.47***	0.057 (0.052–0.061)	0.061 (0.056–0.066)	0.908	–0.016	0.913	–0.014	0.054	0.932	–0.010
Model 3	953.99***	412	30.49	22	18.50***	0.055 (0.051–0.060)	0.060 (0.055–0.065)	0.908	0.000	0.911	–0.002	0.055	0.930	–0.002
Model 4	1022.57***	448	68.34***	36	37.85***	0.055 (0.051–0.059)	0.059 (0.054–0.064)	0.902	–0.006	0.906	–0.005	0.073	0.926	–0.004
Model 6b invariance testing														
Model 0	862.04***	336	–	–	–	0.060 (0.056–0.065)	0.066 (0.060–0.071)	0.911	–	0.915	–	0.051	0.929	–
Model 1	877.32***	350	16.05	14	14.01***	0.059 (0.055–0.064)	0.065 (0.059–0.070)	0.911	0.000	0.915	0.000	0.053	0.928	–0.001
Model 2	968.40***	364	102.01***	14	85.96***	0.062 (0.058–0.067)	0.068 (0.063–0.073)	0.898	–0.013	0.903	–0.012	0.055	0.920	–0.008
Model 3	993.26***	385	30.65	21	27.69***	0.061 (0.056–0.065)	0.066 (0.061–0.072)	0.897	–0.001	0.901	–0.002	0.056	0.918	–0.002
Model 4	1042.90***	413	49.67**	28	37.77***	0.060 (0.056–0.064)	0.065 (0.060–0.070)	0.894	–0.003	0.898	–0.003	0.070	0.915	–0.003
Model 6c invariance testing														
Model 0	676.44***	298	–	–	–	0.054 (0.049–0.059)	0.059 (0.053–0.065)	0.931	–	0.934	–	0.049	0.945	–
Model 1	693.06***	311	16.94	13	13.50***	0.053 (0.049–0.058)	0.058 (0.052–0.064)	0.930	–0.001	0.933	–0.001	0.052	0.944	–0.001
Model 2	781.87***	324	100.13***	13	83.63***	0.057 (0.053–0.062)	0.062 (0.057–0.068)	0.916	–0.015	0.921	–0.012	0.053	0.935	–0.009
Model 3	805.32***	344	28.05	20	22.73***	0.056 (0.051–0.061)	0.061 (0.056–0.067)	0.915	–0.001	0.919	–0.002	0.055	0.933	–0.002
Model 4	857.73***	372	52.36**	28	44.63***	0.055 (0.051–0.060)	0.060 (0.055–0.066)	0.911	–0.004	0.915	–0.004	0.070	0.929	–0.004
Model 6d invariance testing														
Model 0	785.85***	336	–	–	–	0.056 (0.051–0.061)	0.061 (0.055–0.066)	0.923	–	0.927	–	0.050	0.938	–
Model 1	805.07***	350	19.48	14	17.63***	0.055 (0.050–0.060)	0.060 (0.054–0.065)	0.922	–0.001	0.926	–0.001	0.053	0.937	–0.001
Model 2	895.91***	364	101.58***	14	82.67***	0.058 (0.054–0.063)	0.063 (0.058–0.069)	0.909	–0.013	0.914	–0.012	0.054	0.929	–0.008
Model 3	922.84***	385	31.26	21	29.42***	0.057 (0.053–0.061)	0.062 (0.057–0.069)	0.908	–0.001	0.912	–0.002	0.055	0.927	–0.002
Model 4	976.19***	413	53.30**	28	42.11***	0.056 (0.052–0.061)	0.061 (0.057–0.066)	0.904	–0.004	0.908	–0.004	0.069	0.923	–0.004

Table 4 (continued)

CFA model	χ^2 ^a	df	χ^2 ^a	$\Delta\chi^2$ ^a	df	$\Delta\chi^2$ ^a	RMSEA (90% CI) ^b	RMSEA (90% CI) ^c	CFI ^b	ΔCFI^b	CFI ^c	ΔCFI^c	SRMR	GH	ΔGH
Model 10b invariance testing															
Model 0	931.64***	404	–	–	–	–	0.055 (0.051, 0.069)	0.059 (0.054, 0.064)	0.916	–	0.920	–	0.050	0.936	–
Model 1	946.55***	419	16.70	15	15	0.054 (0.050, 0.058)	0.058 (0.053, 0.063)	0.058 (0.053, 0.063)	0.916	0.000	0.919	–0.001	0.053	0.935	–0.001
Model 2	1054.45***	434	119.33***	15	15	0.058 (0.054, 0.062)	0.062 (0.057, 0.067)	0.062 (0.057, 0.067)	0.901	–0.015	0.906	–0.013	0.054	0.926	–0.009
Model 3	1082.23***	457	33.80	23	23	0.056 (0.052, 0.060)	0.061 (0.056, 0.066)	0.061 (0.056, 0.066)	0.900	–0.001	0.904	–0.002	0.055	0.924	–0.002
Model 4	1151.21***	493	68.76***	36	36	0.056 (0.052, 0.060)	0.069 (0.056, 0.065)	0.069 (0.056, 0.065)	0.895	–0.005	0.899	–0.005	0.072	0.920	–0.004

Model 0 = parameters unconstrained across groups. Model 1 = factor loadings constrained equal across groups. Model 2 = factor loadings and item intercepts constrained equal across groups. Model 3 = factor loadings, item intercepts, and residual item variances/covariances constrained equal across groups. Model 4 = item factor loadings, item intercepts, residual item variances/covariances, and factor variances/covariances constrained equal across groups. Individual model chi-square and chi-square difference statistics are robust, using the Satorra and Bentler's (2001) adjustment

CFA confirmatory factor analysis, GH gamma hat

^aChi-square statistic adjusted for non-normality using Satorra and Bentler's (2001) method

^bCalculated using the Satorra and Bentler's (2001) adjusted chi-square statistic

^cCalculated using the methodology outlined in Brosseau-Liard et al. (2012) and/or Brosseau-Liard and Savalei (2014)

* $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$

Table 5 Estimated factor loadings and R^2 s for all items in model 6c for US pre-service and in-service teachers

Item number	Item content	Factor	PST		IST	
			Estimate	R^2	Estimate	R^2
COA15	Assessment provides information on how well schools are doing	Assessment Makes Schools Accountable	0.81	0.65	0.75	0.57
COA22	Assessment is an accurate indicator of a school's quality	Assessment Makes Schools Accountable	0.73	0.53	0.75	0.56
COA27	Assessment is a good way to evaluate a school	Assessment Makes Schools Accountable	0.78	0.61	0.85	0.71
COA7	Assessment feeds back to students their learning needs	Assessment Improves Learning	0.71	0.50	0.69	0.48
COA11	Assessment provides feedback to students about their performance	Assessment Improves Learning	0.79	0.62	0.79	0.63
COA18	Assessment helps students improve their learning	Assessment Improves Learning	0.76	0.57	0.77	0.60
COA4	Assessment information modifies ongoing teaching of students	Assessment Improves Teaching	0.54	0.29	0.68	0.46
COA24	Assessment allows different students to get different instructions	Assessment Improves Teaching	0.68	0.46	0.66	0.43
COA25	Assessment is integrated with teaching practice	Assessment Improves Teaching	0.80	0.64	0.75	0.57
COA10	Assessment results are trustworthy	Assessment Is Valid	0.72	0.52	0.74	0.55
COA12	Assessment results can be depended on	Assessment Is Valid	0.80	0.65	0.88	0.78
COA17	Assessment results are consistent	Assessment Is Valid	0.62	0.39	0.66	0.44
COA8	Assessment interferes with teaching	Assessment Is Bad	0.63	0.39	0.68	0.47
COA19	Assessment is unfair to students	Assessment Is Bad	0.70	0.50	0.63	0.40
COA21	Assessment forces teachers to teach in a way against their beliefs	Assessment Is Bad	0.72	0.52	0.70	0.48
COA3	Assessment results are filed and ignored	Assessment Is Ignored	0.71	0.50	0.72	0.51
COA5	Assessment has little impact on teaching	Assessment Is Ignored	0.46	0.21	0.62	0.38
COA20	Teachers conduct assessments but make little use of the results	Assessment Is Ignored	0.61	0.38	0.63	0.39
COA1	Assessment establishes what students have learned	Assessment Describes Abilities	0.59	0.35	0.60	0.37
COA26	Assessment is a way to determine how much students have learned from teaching	Assessment Describes Abilities	0.82	0.67	0.82	0.67

Estimates are from the single-group confirmatory factor analyses

PST pre-service teacher, IST in-service teacher

Table 6 Correlations among latent variables in model 6c for US pre-service and in-service teachers

Factor	Assessment Is Valid	Assessment Improves Teaching	Assessment Improves Learning	Assessment Makes Schools Accountable	Assessment Is Bad	Assessment Is Ignored	Assessment Describes Abilities
PST							
Assessment Is Valid	1						
Assessment Improves Teaching	0.65	1					
Assessment Improves Learning	0.80	0.74	1				
Assessment Makes Schools Accountable	0.66	0.45	0.63	1			
Assessment Is Bad	-0.44	-0.54	-0.51	-0.33	1		
Assessment Is Ignored	-0.24	-0.39	-0.32	-0.17	0.54	1	
Assessment Describes Abilities	0.69	0.84	0.83	0.64	-0.61	-0.36	1
IST							
Assessment Is Valid	1						
Assessment Improves Teaching	0.72	1					
Assessment Improves Learning	0.79	0.87	1				
Assessment Makes Schools Accountable	0.70	0.50	0.58	1			
Assessment Is Bad	-0.63	-0.72	-0.69	-0.40	1		
Assessment Is Ignored	-0.38	-0.67	-0.45	-0.15	0.76	1	
Assessment Describes Abilities	0.75	0.94	0.79	0.66	-0.71	-0.39	1

Estimates are from single-group confirmatory factor analyses

PST, pre-service teacher; IST, in-service teacher

Table 7 Reliability estimates for model 6c for US pre-service and in-service teachers

Factor	PST			IST		
	Alpha	Omega	Construct replicability	Alpha	Omega	Construct replicability
Assessment Makes Schools Accountable	0.82	0.82	0.82	0.82	0.83	0.84
Assessment Improves Learning	0.80	0.79	0.80	0.79	0.79	0.80
Assessment Improves Teaching	0.70	0.72	0.75	0.74	0.74	0.74
Assessment Is Valid	0.75	0.76	0.78	0.80	0.81	0.84
Assessment Is Bad	0.72	0.73	0.73	0.71	0.71	0.71
Assessment Is Ignored	0.60	0.63	0.65	0.68	0.69	0.70
Assessment Describes Abilities	0.65	0.67	0.72	0.65	0.69	0.72

Estimates are from single-group confirmatory factor analyses

PST pre-service teacher, *IST* in-service teacher

assessment practices (e.g., Reeves et al., 2016; Barnes et al., 2015; Kerr et al., 2006). Despite the importance of teacher beliefs, however, a recent review concluded that teacher beliefs about assessment have, to date, been under-addressed by researchers (Datnow & Hubbard, 2015). At the same time, international research on the validity of inferences derived from most prominent instrument by which to measure teacher beliefs about assessment (the COA-IIIa) has also been mixed, which motivated our study of its score structure among pre-service teachers (PSTs) and in-service teachers (ISTs) in the USA.

Our findings indicate that the original COA-IIIa factor model does not fit well for either US PSTs or ISTs. Most other models we tried either fit poorly as well or were inestimable. However, we were able to identify five relatively good-fitting COA-IIIa models and examine their invariance across these populations: model 5b, model 6b, model 6c, model 6d, and model 10b. In both absolute and relative senses, the best-fitting model for both US PSTs and ISTs was model 6c (Fig. 2). As noted, model 6c featured 20 items and six first-order factors from Brown (2006), but excluded the *Assessment Makes Students Accountable* and *Assessment Is Inaccurate* factors.

Model 6c was effectively a simplification of Brown's (2006) model that removed the higher-order structure, the *Assessment Makes Students Accountable* factor and its three items; removed the *Assessment Is Inaccurate* factor and its three items; and removed one item from the *Assessment Describes Abilities* factor. The item excluded from this factor (COA14, "Assessment measures students' higher-order thinking skills") was also deemed problematic in Daniels et al.'s (2014) study of Canadian pre-service teachers. This item may introduce issues given that it references assessment of higher-order thinking skills specifically, which is certainly not the object of all (or even much) assessment that occurs in US schools.

Our selection here of model 6c as best fitting is based primarily on global fit measures. None of the other models we report here or ran in unreported supplementary analyses fit better globally. That said, model 6c may have some structural issues that need to be sorted out in the future such as the high correlations among

certain factors. That such correlations were especially high among in-service teachers is especially curious and raises the theoretical question of whether a higher-order structure might be appropriate for one population but not the other. One item (COA5, “Assessment has little impact on teaching”) from the *Assessment Is Ignored* factor in model 6c also had a less-than-optimal loading though this was also the case in many of models we closely examined.

The best-fitting models, including model 6c, in our study were not the same as those which fit best in prior COA-III studies, including model 5 in Daniels et al. (2014) (also our model 5) and model C in Brown and Remesal (2012) (our model 8), and the model derived with US in-service teachers by Barnes et al. (2017) (our model 16). However, our best-fitting models did share some similarities with models tested by Daniels et al. (2014) with Canadian PSTs and, in fact, three of the five models that fit best were derived in some way from them. These similarities include dropping the higher-order structure and removal of individual items related to *Assessment Makes Students Accountable*, *Assessment Is Inaccurate*, and *Assessment Describes Abilities* (or removal of these factors entirely).

Our final model was less similar to the so-called India model from Brown et al. (2019), which featured a relatively smaller number of factors, some removed items, and a non-hierarchical structure. This model did not fit well among either US PSTs or ISTs and did not warrant formal invariance testing. As noted by Brown et al. (2019), the simplified features may have contributed to the generalizability of this model cross-nationally, though it does not appear to generalize to the US context. Unlike the “India” model which characterizes beliefs in three main categories, however, our final model does still retain some of the nuance of the original COA-III model.

After an extensive comparison of COA-III model fit among US PST and IST populations, we believe that model 6c may represent a defensible structure to use in the USA. This structure featured fewer items than the original COA-III and thus may be more feasibly implemented due to the smaller number of item and simpler factor structures. However, this structure does sacrifice information about two of Brown’s (2006) original factors and may lack measurement precision for some beliefs. Importantly, the changes made for model 6c relative to Brown’s (2006) model are subtractive only and do not involve relabeling factors, classifying items on different factors, and thus are theoretically defensible.

Cross-cultural differences in COA-III model fit may be potentially explained by methodological and/or theoretical perspectives. Assuming the assessment belief constructs are fixed cross-nationally, differences in findings may be due to cross-national differences in interpretation of items or item wording due to cross-national differences in language, dialects, or discourses around assessment. Again assuming equivalence of the constructs cross-nationally, variance may have also arisen during translations of the instrument (though this would not be the case with our study as English is the dominant language in both New Zealand and the USA).

Alternatively, variation in the COA-III model fit across nations might be due to differences in the underlying belief structures (e.g., presence or absence of different beliefs or different higher-order organizations of beliefs across nations). The findings of Brown et al.’s (2019) multi-national study weigh strongly in favor of this

hypothesis. Varying belief structures may be due to differences across contexts in policies and practices related to assessment (e.g., locus and associated stakes) (Barnes et al., 2015; Daniels et al., 2014), the emphases on different types of assessments (e.g., classroom versus external assessments), or the uses of assessments (e.g., formative versus summative; Leighton et al., 2010; Ruiz-Primo et al., 2002). Our findings do suggest that some of the beliefs posited by Brown (2006) do hold up in the USA, though our best-fitting model does not imply that they are hierarchically structured as originally conceived. From a theoretical perspective, our findings thus suggest both similarities and differences cross-nationally in teachers' underlying belief structures relative to assessment.

None of the best-fitting models, including model 6c, was found to be more than metrically invariant for US in-service and pre-service teachers, which does begs caution with some score uses. This evidence implies that it may be defensible to compare factor variances and covariances across these two groups with these models, but researchers should use caution when deriving other inferences such as factor mean differences. Indeed, our evidence implies that the instrument may not be used effectively interchangeably with these populations. As noted by Barnes et al. (2017) and Brown et al. (2019), COA-III-A users in diverse, and especially novel, contexts should examine factorial structure prior to using its scores to address their research questions about teacher assessment beliefs.

For substantively interested readers, item means differed statistically for 14 of the 20 items in model 6c, with effect sizes (d) generally in the absolute magnitude of 0.10–0.30. PSTs had higher means for one item on the *Assessment Makes Schools Accountable* factor, three items on the *Assessment Improves Learning* factor, two items on the *Assessment Improves Teaching* factor, one item on the *Assessment Is Valid* factor, and two items on the *Assessment Describes Abilities* factor. ISTs had higher means for two items on the *Assessment Is Bad* factor and three items on the *Assessment Is Ignored* factor. Differences on some of these latter items were the largest: “Assessment interferes with teaching” ($d = -0.53$), “Assessment results are filed and ignored” ($d = -0.37$), and “Assessment has little impact on teaching” ($d = -0.32$). Notably, PSTs had higher levels of more positive assessment beliefs, and ISTs had higher levels of more negative assessment beliefs.

5.1 Limitations, delimitations, and future directions

Our findings should certainly be interpreted in light of our non-random sampling design for both the PST and IST samples and, also, the inclusion of only Illinois ISTs. However, we re-analyzed our data including only Illinois PSTs and results were very similar such that our substantive findings about model fit and invariance would not have changed. We also encountered sporadic estimation issues which may indicate issues with the data themselves, rather than model fit issues. Our substantive and analytic judgments may have also masqueraded more optimal structures. We collegially invite other studies to independently validate our

findings. Future researchers may also consider the use of modification indices (Jung & Yoon, 2016) or examination of partial invariance in independent samples and the collection of other forms of validity evidence. Given the large number of models we tested, the substantive and empirical judgments we made, and our non-probabilistic sample, we opted not to use modification indices or conduct partial scalar invariance testing.

A theoretical matter also deserves future attention. Model 6c did not incorporate the *Assessment Makes Students Accountable* factor. This finding could imply that this is not a distinct belief held by US PSTs or ISTs, which may be explained by the stakes associated with assessment in the USA. While nations such as Hong Kong and the USA are often thought of as having higher stakes associated with assessment (Barnes et al., 2015; Daniels et al., 2014), in the USA, these stakes are most often associated with schools (or teachers) rather than students. Indeed, accountability provisions in the federal No Child Left Behind Act and its successor, the Every Student Succeeds Act, generally target schools. On the other hand, this finding could be due to the idiosyncratic content of those specific items used in the COA-III (e.g., “qualifications standards”) to operationalize this belief. At the same time, the removal of the *Assessment Is Inaccurate* factor and its items related to measurement error and precision may imply that US PST and IST beliefs do not hold a distinct belief about the technical soundness of assessment. Our findings thus beg further theoretical or measurement research concerning these beliefs in the USA.

Acknowledgements The authors thank all pre-service and in-service teachers who participated in the study, as well as Amy Martin, Evelyn Comber, David Walker, Kelly Summers, and Thomas Smith for their feedback provided on an earlier version of this manuscript. We also thank Gavin Brown for his swift and helpful assistance in specifying several of the tested models, and Mary Sanderson for the data analytic support.

Data availability The data that support the findings of this study are available from the authors upon reasonable request.

Declarations

Conflict of interest The authors declare no competing interests.

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